

Question ID: 8f82ad81

| Assessment | Test | Domain        | Skill                  | Difficulty                                        |
|------------|------|---------------|------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Easy <div><div></div><div></div><div></div></div> |

Question

Which expression is equivalent to  $4(x^2 + 6)$ ?

Answer

- A.  $4x^2 + 24$
- B.  $4x^2 + 10$
- C.  $4x^2 + 6$
- D.  $4x^2 - 2$

Correct Answer: A

Rationale

Choice A is correct. The expression  $4(x^2 + 6)$  can be rewritten as  $4(x^2) + 4(6)$ , which is equivalent to  $4x^2 + 24$ .

Choice B is incorrect. This expression is equivalent to  $4(x^2 + \frac{5}{2})$ , not  $4(x^2 + 6)$ .

Choice C is incorrect. This expression is equivalent to  $4(x^2 + \frac{3}{2})$ , not  $4(x^2 + 6)$ .

Choice D is incorrect. This expression is equivalent to  $4(x^2 - \frac{1}{2})$ , not  $4(x^2 + 6)$ .